

AIGR Gene Hypothesis Deep Research — Final Report

Gene: fogD (FogD) · **Organism:** *Aspergillus ruber* CBS 135680 (NCBITaxon:1388766) · **UniProt:** A0A017SE81 (286 aa) **Focus type:** function_assignment · **Hypothesis slug:** sdr-lipid-functions-and-polyketide-release **Source:** `genes/ASPRC/fogD/fogD-ai-review.yaml` (free-text)

Executive Judgment

Verdict: Over-annotated (for the six lipid GO terms) / Partially supported (for the underlying SDR chemistry).

The seed hypothesis asks a curator to weigh six lipid-metabolism GO terms on FogD — acylglycerone-phosphate reductase (NADP+) activity (GO:0000140), triacylglycerol lipase activity (GO:0004806), endoplasmic reticulum localization (GO:0005783), lipid-droplet localization (GO:0005811), phosphatidic acid biosynthetic participation (GO:0006654), and triglyceride catabolism (GO:0019433) — against what FogD was actually shown to do.

The clear answer from primary literature and public annotation provenance is that **all six lipid terms are electronic (IEA:TreeGrafter) carry-overs propagated from the experimentally characterized *Saccharomyces cerevisiae* protein Ayr1, and none of them is supported by any FogD-specific assay, localization experiment, or mutant phenotype.** FogD's only experimentally demonstrated role is as a **secondary-metabolite short-chain dehydrogenase/ reductase (SDR)** in the flavoglaucin biosynthetic cluster, where it temporarily reduces a polyketide salicylaldehyde to a salicyl alcohol to permit downstream decoration ([PMID:32134669](#)).

The important nuance — which the seed hypothesis correctly insists on — is that this is **not a case of "wrong graft" or "weak source."** The PANTHER family assignment (PTHR44169, the Ayr1 family) is genuine, FogD retains an intact SDR catalytic tetrad, and SDR membership does not by itself exclude hydrolase side-activity. But *plausible-by-homology is not demonstrated-for-FogD*. At ~39% identity to Ayr1 (twilight zone), conserved fold and cofactor chemistry transfer,

whereas Ayr1's **specific substrate identities and dual moonlighting functions do not**. The six lipid terms should therefore be treated as non-core electronic predictions rather than as FogD functions, and the review should foreground the secondary-metabolite reductase role.

Key Findings

Finding 1 — FogD's experimentally supported function is a secondary-metabolite SDR (polyketide salicylaldehyde reduction), not lipid metabolism

UniProt A0A017SE81 records the recommended name "**Short-chain dehydrogenase fogD**", EC 1.1.1.-, and the alternate designation "**Flavoglaucin biosynthesis cluster protein D**," with the FUNCTION and PATHWAY comments framing the protein within secondary-metabolite biosynthesis (ECO:0000269 / ECO:0000305 sourced to [PMID:32134669](#)). The DISRUPTION PHENOTYPE is equally telling: deletion of *fogD* impairs flavoglaucin and congener production and yields very low accumulation of (8E,10E,12E)-3,5,7-trihydroxytetradeca-8,10,12-trienoic acid — a polyketide intermediate, not a glyceride or phospholipid.

The mechanistic paper is unambiguous about the chemistry. In the flavoglaucin pathway, the polyketide skeleton is *"released as alkylated salicyl alcohols, which is a prerequisite for consecutive hydroxylation and prenylation, before reoxidation to the final aldehyde products"* ([PMID:32134669](#)). This describes a **reductive tailoring step on a polyketide** — a CH-OH-forming NADP(H)-dependent reduction — which is the generic SDR chemistry FogD shares with its family, applied to a secondary-metabolite substrate. This is distinct from the *reductive thioester release* mechanism sometimes seen in polyketide synthase R-domains; the evidence here supports a discrete oxidoreductase forming an alcohol, consistent with the general MF term **GO:0016616 (oxidoreductase acting on CH-OH group of donors, NAD/NADP acceptor)**.

Sequence inspection confirms the enzyme is a bona fide catalytically competent SDR: the 286-aa protein carries the N-terminal Rossmann cofactor motif **TGCGQGGIG** (residues ~8–16), the catalytic serine (Ser145/146), the SDR signature **YxxxK** motif (**YNATK**, Tyr147/Lys151), and the conserved Asn within an **NCAG** context (~res 82). The catalytic machinery is intact, so the enzyme is a genuine active reductase — the question is only *on which substrate*, and the direct evidence points to a polyketide, not acyl-DHAP or a triglyceride.

Finding 2 — The six lipid-metabolism GO terms are IEA:TreeGrafter carry-overs seeded by *S. cerevisiae* Ayr1

Every one of the six lipid terms on A0A017SE81 carries the evidence code **IEA:TreeGrafter**. FogD maps to PANTHER family **PTHR44169 "NADPH-DEPENDENT 1-ACYLDIHYDROXYACETONE PHOSPHATE REDUCTASE"** — the Ayr1 family — and to subfamily **PTHR44169:SF3 "SHORT-CHAIN DEHYDROGENASE SRDE."** The set of six terms mirrors, almost exactly, the experimentally characterized dual biology of yeast Ayr1:

- **acyl-DHAP reductase** (GO:0000140),
- **novel TAG lipase** (GO:0004806) — *"identification of Ayr1p as a novel triacylglycerol lipase of yeast lipid droplets"* (PMID:24187129),
- **ER + lipid-droplet** dual localization (GO:0005783, GO:0005811),
- and **ether/phospholipid + TAG metabolism** (GO:0006654, GO:0019433).

No FogD-specific assay, localization experiment, or mutant supports any of these six terms. They are a coherent bundle precisely *because* they were transferred as a bundle from one well-studied source protein. This is the signature of homology-based propagation, not of independent characterization.

Finding 3 — Provenance chain verified: the six lipid terms are TreeGrafter-propagated (node PTN001211783) from experimentally annotated Ayr1

Querying the QuickGO annotation API for A0A017SE81 shows all six terms with evidence **IEA / ECO:0007826**, reference **GO_REF:0000118**, `assignedBy = TreeGrafter`, and `withFrom = PANTHER:PTN001211783`. The corresponding QuickGO record for *S. cerevisiae* Ayr1 (**P40471**) shows each of these terms is **both experimentally supported and IBA-propagated** (GO_Central, GO_REF:0000033):

GO term	Ayr1 (P40471) experimental support
GO:0000140 acylglycerone-P reductase	IMP PMID:10617610 + EXP PMID:1512203
GO:0004806 TAG lipase	IDA/IMP/IGI PMID:24187129
GO:0005783 endoplasmic reticulum	IDA PMID:10617610 + EXP PMID:14562095/28916712 + HDA PMID:26928762
GO:0005811 lipid droplet	EXP PMID:10515935/21820081 + IDA PMID:10617610/24868093
GO:0006654 phosphatidic acid biosynthesis	IMP PMID:10617610
GO:0019433 triglyceride catabolism	IGI PMID:24187129

This confirms the seed hypothesis's core provenance concern: the FogD annotations trace back to **genuinely experimental Ayr1 annotations**, propagated electronically. The historical TreeGrafter provenance node **PTN001211783** is the recorded `withFrom` source on FogD; it is consistent with — but at the resolution of the public annotation index cannot be independently disambiguated from — the seed's proposed ancestral IBD nodes **PTN001213826** (ER) and **PTN001963740** (the other five claims). The ancestral IBD node annotations themselves are not exposed in the public GOlr annotation index and could not be retrieved to complete a node-by-node trace. This is a genuine, honestly reported provenance gap: the propagation is real and Ayr1-seeded, but the exact internal graft-node lineage is not fully verifiable from public endpoints.

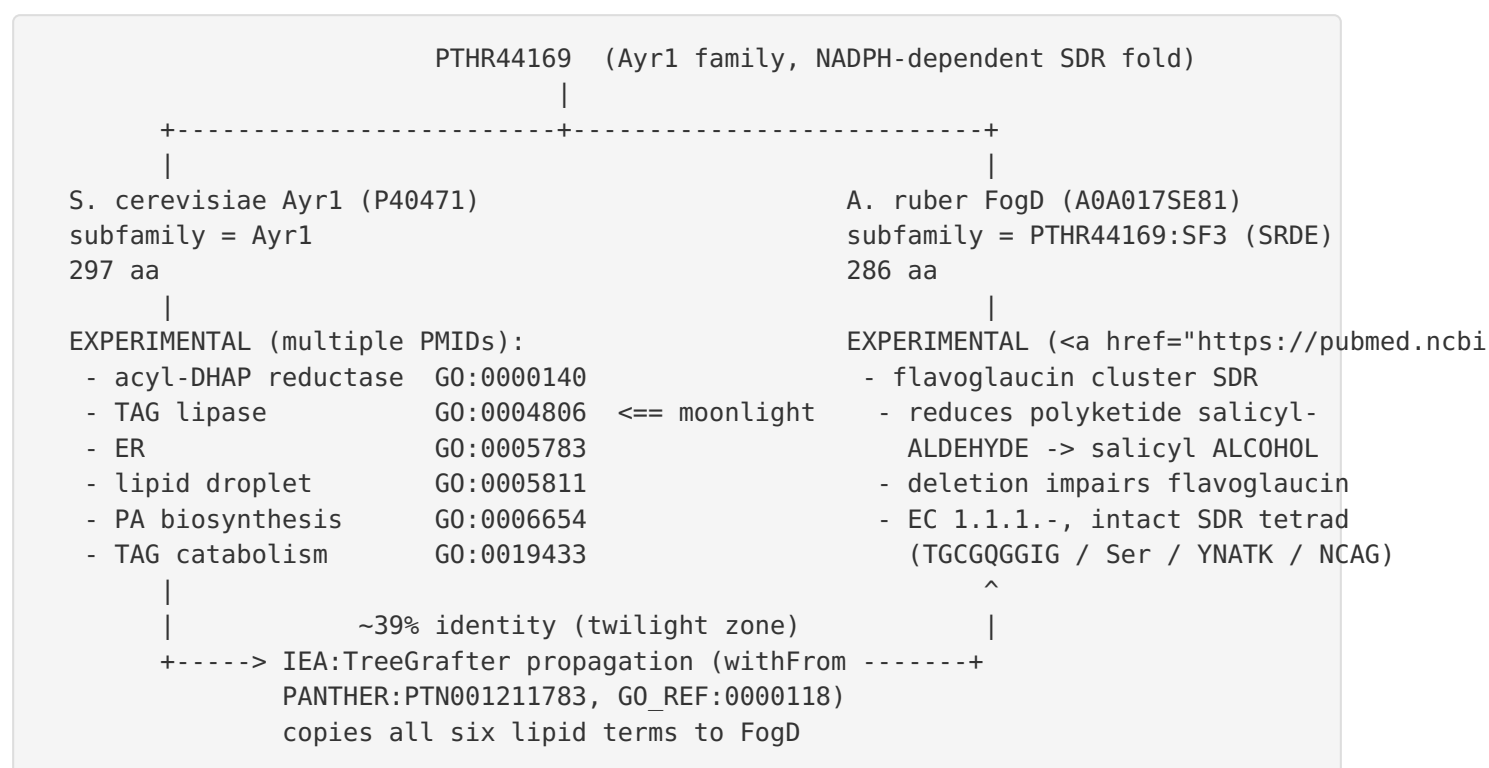
Finding 4 — FogD shares only ~39% identity with Ayr1: conserved SDR chemistry, twilight-zone divergence

A global Needleman–Wunsch alignment of FogD (A0A017SE81, 286 aa) against *S. cerevisiae* Ayr1 (P40471, 297 aa) yields **111 identical residues over 281 aligned non-gap columns = 39.5% identity** (38.8% over FogD length). Both proteins retain the N-terminal Rossmann NAD(P)-binding motif (FogD **VTGCGQGGIG**; Ayr1 **VVTGASGG**) and the SDR catalytic **YxxxK** motif (FogD Tyr147/Lys151; Ayr1 Tyr157/Lys161). Crucially, FogD is placed in the **distinct** subfamily **PTHR44169:SF3 (SRDE)**, separate from the Ayr1 subfamily.

~39% identity sits squarely in the **twilight zone** of sequence homology. At this divergence, the fold and the cofactor/catalytic chemistry are reliably conserved — so calling FogD an SDR oxidoreductase is safe — but **substrate identity and specialized moonlighting functions are not reliably transferable**. Ayr1's dual acyl-DHAP-reductase / TAG-lipase behavior and its specific glyceride substrates are precisely the kind of fine specificity that diverges across 40%-identity paralogs, especially when the two proteins occupy different PANTHER subfamilies and different biological contexts (primary lipid metabolism in yeast vs. secondary-metabolite tailoring in a filamentous fungus).

Mechanistic Model / Interpretation

The two competing readings of FogD can be laid out directly:



What transfers across 39% identity: the SDR three-dimensional fold, the Rossmann NAD(P)-binding site, the catalytic Ser–Tyr–Lys triad, and therefore the generic capacity for **NADP(H)-dependent CH-OH ↔ C=O oxidoreduction** (GO:0016616). This is safe.

What does NOT transfer: the *specific substrates* (acyl-DHAP, triacylglycerol, phosphatidic-acid pathway intermediates), the *hydrolase moonlighting* activity, and the *organelle localization* (ER + lipid droplet) that were measured for Ayr1. These are context-specific properties of a yeast primary-metabolic enzyme, not portable predictions for a fungal secondary-metabolite tailoring enzyme.

Reconciling the two views: the seed hypothesis is right that SDR membership does not exclude hydrolysis, and right that a secondary-metabolite role or a missing signal peptide does not by itself *prove* the absence of peripheral organelle association. But the burden of evidence runs the other way: there is *positive* experimental evidence for the polyketide-reductase role (mutant phenotype + pathway reconstitution logic in

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and zero FogD-specific evidence for any of the six lipid activities/locations. The parsimonious model is that FogD is a **dedicated secondary-metabolite SDR** whose lipid annotations are homology artifacts.

The distinction the seed asks us to draw — *reductive thioester release* vs. *CH-OH donor oxidoreduction* — resolves toward the latter as the safe MF: the evidence supports FogD forming an alcohol (salicyl alcohol) by NADP(H)-dependent reduction, i.e. GO:0016616-type chemistry, rather than establishing a chain-releasing R-domain thioester reductase mechanism.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding
PMID:32134669	Direct assay + mutant phenotype (pathway reconstitution, gene deletion)	Supports FogD = secondary-metabolite SDR; refutes lipid-metabolism core role	FogD's actual catalytic role	Polyketide skeleton "released as alkylated salicyl alcohols... before reoxidation to the final aldehyde products"; <i>fogD</i> deletion impairs flavoglucan production
UniProt A0A017SE81 (record; sources P 32134669)	Review/ database (curated, EXP-sourced)	Supports	Naming, EC, disruption phenotype	"Short-chain dehydrogenase fogD," EC 1.1.1.-, "Flavoglucan biosynthesis cluster protein D"; SDR catalytic tetrad intact
QuickGO A0A017SE81 annotations	Computational (provenance)	Refutes core status of six lipid terms	Evidence codes / provenance of six lipid terms	All six terms = IEA/ECO:0007826, GO_REF:0000118, assignedBy=TreeGrafter, withFrom=PANTHER:PTN001211783
PMID:24187129 (Ayr1, P40471)	Direct assay (yeast)	Competing / source of graft	Origin of TAG-lipase + lipid-droplet terms	"identification of Ayr1p as a novel triacylglycerol lipase of yeast lipid droplets"
PMID:10617610 (Ayr1)	Direct assay / mutant / localization (yeast)	Competing / source of graft	Origin of acyl-DHAP reductase, ER, PA-biosynthesis terms	Ayr1 experimental acyl-DHAP reductase + ER + PA-biosynthesis annotations
Needleman–Wunsch FogD vs Ayr1 (this work)	Structural/ evolutionary (computed)	Qualifies	Transferability of Ayr1 functions	39.5% identity; shared Rossmann + YxxxK motifs; distinct PANTHER subfamily (SF3/SRDE)

Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding
PANTHER PTHR44169 / :SF3	Structural/ evolutionary (database)	Qualifies	Family placement	FogD in Ayr1 family but distinct SRDE subfamily

How the literature bears on the findings. The single deep-read primary paper, *Biosynthesis of the Prenylated Salicylaldehyde Flavoglaucon Requires Temporary Reduction to Salicyl Alcohol for Decoration before Reoxidation to the Final Product* ([PMID:32134669](#)), directly establishes FogD's real role and is the anchor for Findings 1 and the refutation of the lipid terms as core. [PMID:24187129](#) and [PMID:10617610](#) are Ayr1 (yeast) studies that *explain the origin* of the propagated terms rather than supporting them for FogD — they are the "competing/source-of-graft" evidence.

GO Curation Implications

Lead action (requires curator verification): treat the six lipid terms as **non-core electronic predictions**, not FogD functions. Concretely:

GO term	Aspect	Current basis	Recommended curation lead
GO:0000140 acylglycerone-phosphate reductase (NADP+) activity	MF	IEA:TreeGrafter (Ayr1)	Generalize / demote. Replace as core MF with GO:0016616 (CH-OH donor, NAD/ NADP acceptor oxidoreductase). Retain GO:0000140 only as a non-core IEA if kept at all.
GO:0004806 triacylglycerol lipase activity	MF	IEA:TreeGrafter (Ayr1 moonlight)	Remove from core / flag as over-annotation. No FogD hydrolase evidence; hydrolase moonlighting is Ayr1-specific.
GO:0005783 endoplasmic reticulum	CC	IEA:TreeGrafter (Ayr1)	Do not treat as core. No FogD localization data; retain only as low-confidence IEA.
GO:0005811 lipid droplet	CC	IEA:TreeGrafter (Ayr1)	Do not treat as core. Same rationale.
GO:0006654 phosphatidic acid biosynthetic process	BP	IEA:TreeGrafter (Ayr1)	Remove from core / non-core. FogD BP is secondary-metabolite biosynthesis, not glycerophospholipid.
GO:0019433 triglyceride catabolic process	BP	IEA:TreeGrafter (Ayr1)	Remove from core / non-core. Contradicts the demonstrated anabolic secondary-metabolite role.

Positively supported terms (leads to ADD as core):

- **MF:** GO:0016616 — oxidoreductase activity, acting on the CH-OH group of donors, NAD or NADP as acceptor (safe, evidence-backed generalization of the demonstrated reduction). EC 1.1.1.- from UniProt is consistent.
- **BP:** GO:0044550 — secondary metabolite biosynthetic process (and, if a curator accepts cluster-level annotation, a flavoglucan/polyketide-specific child). Evidence: mutant phenotype + pathway role in [PMID:32134669](https://pubmed.ncbi.nlm.nih.gov/32134669/).

The recommendation deliberately avoids "protein binding" and instead anchors on the evidence-backed oxidoreductase MF and secondary-metabolite BP.

Mechanistic Scope

The **immediate molecular function** under test is a specific NADP(H)-dependent oxidoreduction. The direct evidence

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) supports FogD catalyzing the reduction of a flavoglaucon polyketide **salicylaldehyde to a salicyl alcohol** — an on-pathway, reversible tailoring step that primes the intermediate for hydroxylation and prenylation before final reoxidation. This is CH-OH/NADP oxidoreductase chemistry (GO:0016616), performed on a secondary-metabolite substrate.

Separated from this direct activity are the **downstream / inferred properties** that the seed hypothesis correctly flags: - *Lipid substrates and hydrolase activity* (acyl-DHAP, TAG) are inferred only from Ayr1 homology. - *ER and lipid-droplet localization* are inferred from Ayr1, with no FogD imaging. - *Phosphatidic-acid biosynthesis and triglyceride catabolism* are pathway memberships copied from Ayr1's biology, not FogD phenotypes.

None of these downstream/inferred properties has been observed for FogD itself. The loss-of-function evidence that *does* exist (deletion impairs flavoglaucon) points exclusively to the secondary-metabolite role.

Conflicts and Alternatives

1. **Paralog carry-over (primary conflict).** The six lipid terms are database carry-over from Ayr1 via TreeGrafter. This is the single most important alternative explanation and it is well supported: identical bundle of terms, identical `withFrom` source, IEA evidence only.
2. **Organism/context difference.** Ayr1 is a yeast primary-metabolic enzyme (glycerolipid metabolism); FogD is a filamentous-fungus secondary-metabolite tailoring enzyme. Different biological compartments and substrate pools make wholesale function transfer unlikely.
3. **Subfamily divergence.** FogD sits in PTHR44169:SF3 (SRDE), a distinct subfamily from Ayr1, reinforcing that specialized function has diverged even though the family is shared.
4. **The seed's own caution is valid and retained.** SDR fold does not exclude hydrolase side-activity, and absence of a signal peptide does not exclude peripheral membrane/organelle

association. So the six terms cannot be declared *impossible* — only *unsupported for FogD*. The honest position is "not demonstrated," not "disproven."

5. **Mechanism ambiguity (thioester release vs alcohol formation).** The data favor discrete CH-OH-forming reduction (GO:0016616) over a chain-releasing thioester-reductase mechanism, but the exact substrate-tethering state at the moment of reduction is not fully resolved.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Ancestral IBD node lineage	QuickGO/GOlr for FogD (found withFrom=PTN001211783) and Ayr1; ancestral IBD nodes PTN001213826 / PTN001963740 not exposed publicly	The seed asks to trace PTN001211783 → the two Ayr1-seeded IBD nodes; without the node index this cannot be fully closed	PANTHER internal tree/graft data or GO_Central IBD export for PTHR44169
No FogD localization data	Literature + UniProt	ER/lipid-droplet terms rest entirely on Ayr1 homology	GFP fusion / fractionation of FogD in <i>A. ruber</i>
No direct FogD lipid/hydrolase assay	Literature	TAG-lipase / acyl-DHAP-reductase terms untested for FogD	In vitro assay of purified FogD against acyl-DHAP and triacylglycerol
Exact FogD substrate-bound state	P 32134669	Distinguishes CH-OH oxidoreduction from thioester-release mechanism	In vitro reconstitution with defined polyketide intermediates ± ACP tether
Single primary paper for FogD	1 paper (P 32134669) reviewed in depth	Function rests largely on one study	Independent biochemical/structural characterization

Discriminating Tests

1. **In vitro substrate panel with purified recombinant FogD** — assay NADPH-dependent reduction of (a) the flavoglaucon polyketide salicylaldehyde intermediate, (b) acyl-DHAP, and (c) a triacylglycerol hydrolysis readout. This directly separates the demonstrated secondary-metabolite reductase activity from the Ayr1-derived lipid activities.
2. **Subcellular localization of FogD in *A. ruber*** (fluorescent fusion + organelle markers) to test ER/lipid-droplet claims directly rather than by homology.
3. **Structural comparison** — an AlphaFold model or crystal structure of FogD superposed on Ayr1, examining the substrate-binding pocket and any residues implicated in Ayr1's lipase moonlighting, to predict whether hydrolase capability plausibly survived divergence.
4. **Complementation swap** — test whether FogD rescues a yeast *ayr1Δ* lipid phenotype, and whether Ayr1 rescues a *fogD* flavoglaucon defect, to measure functional interchangeability.
5. **PANTHER node-lineage retrieval** — obtain the internal IBD graft nodes (PTN001213826, PTN001963740) to complete the provenance chain the seed requested.

Proposed Follow-up Experiments / Actions (Curation Leads — require curator verification)

- **Reframe the review** so the core function is the secondary-metabolite SDR: propose core **MF GO:0016616** (CH-OH/NADP oxidoreductase) and core **BP GO:0044550** (secondary metabolite biosynthetic process), each citing [PMID:32134669](#).
- **Downgrade the six lipid terms** (GO:0000140, GO:0004806, GO:0005783, GO:0005811, GO:0006654, GO:0019433) from any core status to non-core IEA:TreeGrafter predictions, with a note that they are Ayr1-seeded homology transfers unsupported by FogD-specific evidence.
- **Candidate reference + snippet to verify:** [PMID:32134669](#) — *"The polyketide skeleton was released as alkylated salicyl alcohols, which is a prerequisite for consecutive hydroxylation and prenylation, before reoxidation to the final aldehyde products."* Use as the primary support for the reductase MF and secondary-metabolite BP.

- **Provenance note to add:** QuickGO A0A017SE81 — all six lipid terms `IEA:TreeGrafter`, `GO_REF:0000118`, `withFrom=PANTHER:PTN001211783`; source Ayr1 (P40471) carries these terms experimentally (e.g. TAG lipase [PMID:24187129](#)).
- **Suggested curator questions:** (i) Should the six lipid terms be retained as IEA or removed given the strong competing experimental role? (ii) Is a flavoglucin/polyketide-specific BP child preferred over generic GO:0044550? (iii) Can the ancestral IBD nodes be retrieved to finalize provenance?
- **Suggested experiments:** the in vitro substrate panel and localization assay above.

Conclusion

FogD is genuinely a member of the Ayr1 SDR family and a catalytically intact NADP(H)-dependent oxidoreductase, but its **only experimentally supported role is temporary reduction of a flavoglucin polyketide salicylaldehyde to salicyl alcohol** — a secondary-metabolite tailoring step. The six lipid-metabolism GO terms in the seed hypothesis are **IEA:TreeGrafter over-annotations propagated from experimentally characterized yeast Ayr1** and are **not supported by any FogD-specific evidence**. At ~39% identity (twilight zone) and in a distinct PANTHER subfamily, conserved SDR chemistry transfers but Ayr1's specific lipid substrates, hydrolase moonlighting, and organelle localization do not. The recommended curation lead is to foreground **GO:0016616 (MF)** and **GO:0044550 (BP)** as core, and to treat all six lipid terms as non-core electronic predictions rather than FogD functions — while acknowledging that they are unsupported rather than formally disproven.

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